



EDUFYI TECH SOLUTIONS

BIOTECHNOLOGY ENGINEERING

Advanced Certification Program • 2026 Edition

SOLO SPRINT

Self-paced

LIVE EDGE

Mentor-led

CAREER EDGE

Job-ready

GENE EDITING • BIOPROCESSING • BIOINFORMATICS • SYNTHETIC BIOLOGY • AI FOR BIOLOGY

12 weeks | 18 wet-lab & dry-lab modules | 10 frontier electives | industry capstone

12

Weeks

28

Lab & lab-code modules

10

Frontier electives

100%

Placement support

Built by practitioners

Mentors from biopharma R&D, CDMOs and research labs.

Wet lab + dry lab

Bench work and computational biology in one track.

Portfolio that hires

A defended capstone, code and lab notebooks you own.



Why Biotechnology, Why Now

Biotechnology is where biology becomes engineering. This programme takes you from molecular fundamentals to industrial-grade gene editing, bioprocess scale-up and AI-driven drug discovery in twelve intensive weeks — built and mentored by working scientists.

\$1.6T

Global bio-economy by 2030

14%

CAGR, Indian biotech sector

70K+

New biotech roles a year

6

Career tracks covered

WHAT YOU WILL MASTER

- CRISPR-Cas9, base & prime editing workflows
- Recombinant protein expression and purification
- Upstream & downstream bioprocess scale-up
- NGS data analysis and genomic variant calling
- Synthetic biology circuit design (SBOL, iGEM style)
- mAb, vaccine and cell-therapy manufacturing
- AI/ML for target discovery and lead optimisation
- Regulatory affairs: CDSCO, USFDA, GxP, biosafety
- Bioreactor operation and downstream purification
- AlphaFold structure prediction and molecular docking
- Scientific writing, SOPs and data integrity (ALCOA+)
- Bio-IPR, patents and technology transfer basics

WHO IT IS FOR

- B.Sc / M.Sc Biotech, Microbiology, Biochemistry
- B.Tech Biotech & Bioinformatics students
- Pharma / QC / QA professionals upskilling
- Life-science graduates entering R&D
- Researchers moving into industry roles
- Career switchers with a science background

PREREQUISITE

Basic biology at 12th-grade level. No prior coding experience required — Python is taught from first principles.

HOW YOU LEARN

01

60% hands-on lab and computational work

02

Weekly graded assignments with mentor review

03

One capstone defended before an industry panel

Dual outcome: an industry-validated certificate plus a documented research portfolio.

Every learner ships a reproducible capstone with real data, code and lab notebooks.



01 Molecular & Cellular Foundations

WEEKS 1-2

- Cell architecture, membrane dynamics, organelle bioenergetics
- DNA/RNA chemistry, replication, transcription, translation control
- Enzyme kinetics, protein folding, structure–function relationships
- Aseptic technique, microbial culture, sterilisation & biosafety levels
- Lab: micropipetting, media prep, gram staining, growth-curve assay

PHASE OUTCOME **Confident, contamination-free bench work**

02 Genetic Engineering & Molecular Tools

WEEKS 3-4

- Restriction cloning, Gibson assembly, Golden Gate & vector design
- PCR, qPCR, ddPCR, primer design and troubleshooting
- Gel electrophoresis, Sanger sequencing, plasmid QC
- CRISPR-Cas9, base editing, prime editing, guide-RNA design
- Lab: transform E. coli, screen colonies, validate an edit

PHASE OUTCOME **A validated, sequence-verified construct**

03 Bioprocess & Industrial Biotechnology

WEEKS 5-6

- Upstream: strain development, media optimisation, fed-batch vs perfusion
- Bioreactor design, DO/pH control, scale-up from 5 L to 2000 L
- Downstream: cell lysis, chromatography, TFF, lyophilisation
- Process analytics, yield economics, single-use technology
- Lab: fermentation run with real-time parameter logging

PHASE OUTCOME **A documented, costed fermentation run**



04 Bioinformatics & Computational Biology

WEEKS 7-8

- Python & R for biological data; Biopython, pandas, ggplot
- NGS pipelines: QC, alignment, variant calling, RNA-seq DE analysis
- Protein structure prediction with AlphaFold; molecular docking
- Machine learning for drug discovery and biomarker prediction
- Lab: end-to-end RNA-seq analysis on a public dataset

PHASE OUTCOME **A reproducible analysis notebook on GitHub**

05 Advanced & Applied Biotechnology

WEEKS 9-10

- Synthetic biology: genetic circuits, biosensors, metabolic engineering
- Monoclonal antibodies, vaccine platforms (mRNA, viral vector, subunit)
- Cell & gene therapy: CAR-T, AAV vectors, organoids, stem-cell basics
- Agri & environmental biotech: transgenics, biofuels, bioremediation
- Nanobiotechnology, lab-on-chip diagnostics, biosensing devices

PHASE OUTCOME **A chosen specialisation and research brief**

06 Regulatory, Quality & Career Launch

WEEKS 11-12

- GLP/GMP, ISO 13485, CDSCO & USFDA submission pathways
- Bio-IPR, patents, tech transfer and biotech entrepreneurship
- Scientific writing, SOPs, batch records and data integrity (ALCOA+)
- Portfolio, resume, domain + HR mock interviews
- Capstone defence before an industry mentor panel

PHASE OUTCOME **Job-ready portfolio and interview readiness**



Advanced Topics & Toolchain

Beyond the core track, every cohort runs six frontier modules that mirror what R&D teams are actually hiring for in 2026.



AI-Driven Drug Discovery

Generative chemistry, ADMET prediction, active-learning virtual screening pipelines.

MODULE 01



Single-Cell & Spatial Multi-Omics

scRNA-seq, ATAC-seq, spatial transcriptomics; Seurat & Scanpy workflows.

MODULE 02



mRNA & LNP Platforms

IVT design, cap analogues, lipid nanoparticle formulation and cold-chain logistics.

MODULE 03



Synthetic Genomics

Genome minimisation, xeno-nucleic acids, biocontainment and kill-switch design.

MODULE 04



Precision Fermentation

Cultivated protein, mycoprotein, techno-economic modelling for food biotech.

MODULE 05



Bioethics & Biosecurity

Dual-use research, germline governance, ethical review and global frameworks.

MODULE 06

TOOLCHAIN YOU WILL USE

Benchling

SnapGene

Geneious

AlphaFold 3

PyMOL

AutoDock

RFdiffusion

ESMFold

Python / Biopython

R / Bioconductor

Seurat

Scanpy

Galaxy

GROMACS

SBOL Designer

Cell Designer

GraphPad Prism

ImageJ / Fiji

Nextflow

Docker

Git / GitHub

SuperPro Designer

Hugging Face

Weights & Biases

Every tool is taught hands-on with licensed or open-source access provided for the cohort.



The 2026 Advanced Track

Pick any two deep-dive electives to run alongside weeks 7–12. Each is a four-session sprint with its own mini-project, taught by a specialist practitioner.

ELECTIVE 01

Protein Design with Generative AI

RFdiffusion, ProteinMPNN and ESM-2 embeddings to design de-novo binders, then validate them in silico with MD.

- De-novo binder & enzyme scaffolds
- Inverse folding and sequence recovery
- MD stability screening in GROMACS

ELECTIVE 02

CRISPR 3.0 — Epigenome & RNA Editing

Beyond cutting: CRISPRa/i, dCas9 effectors, prime-editing v3, ADAR-based RNA editors and delivery strategy.

- Epigenetic writers and erasers
- Prime editor architecture & pegRNA
- LNP / AAV delivery trade-offs

ELECTIVE 03

Organ-on-Chip & Advanced 3D Models

Microfluidic tissue chips, organoids and bioprinting as replacements for animal models under the FDA Modernization Act.

- Chip design and perfusion
- Organoid maturation protocols
- Toxicology read-outs and imaging

ELECTIVE 04

Biomanufacturing 4.0 & Digital Twins

Mechanistic + ML hybrid models, PAT sensors, MPC control and QbD for continuous, self-optimising bioprocess plants.

- Kinetic modelling and soft sensors
- Real-time release & PAT
- Techno-economic simulation

ELECTIVE 05

Lab Automation & Self-Driving Labs

Opentrons/liquid handlers, robotic scheduling, Bayesian design of experiments and closed-loop autonomous discovery.

- Protocol-as-code (Python API)
- Active learning DoE loops
- LIMS and FAIR data capture

ELECTIVE 06

Microbiome & Metagenomic Engineering

Community profiling, metabolic modelling of consortia and engineered live biotherapeutics.

- Shotgun metagenomics pipelines
- Genome-scale models (COBRApy)
- Live biotherapeutic regulation

ELECTIVE 07

Quantum & HPC Molecular Simulation

Free-energy perturbation, GPU-accelerated MD and early quantum-computing methods for binding-affinity prediction.

- FEP/TI binding free energies
- GPU & cluster job orchestration
- Variational quantum chemistry primer

ELECTIVE 08

Biosecurity, Provenance & Bio-AI Governance

Sequence screening, dual-use risk assessment, model release policy and the emerging AI-bio regulatory perimeter.

- Automated sequence screening
- Red-teaming bio-AI models
- Global governance frameworks

RESEARCH PATHWAY

Top performers are invited to extend an elective mini-project into a publishable preprint or a provisional patent filing, supported by our research mentors.

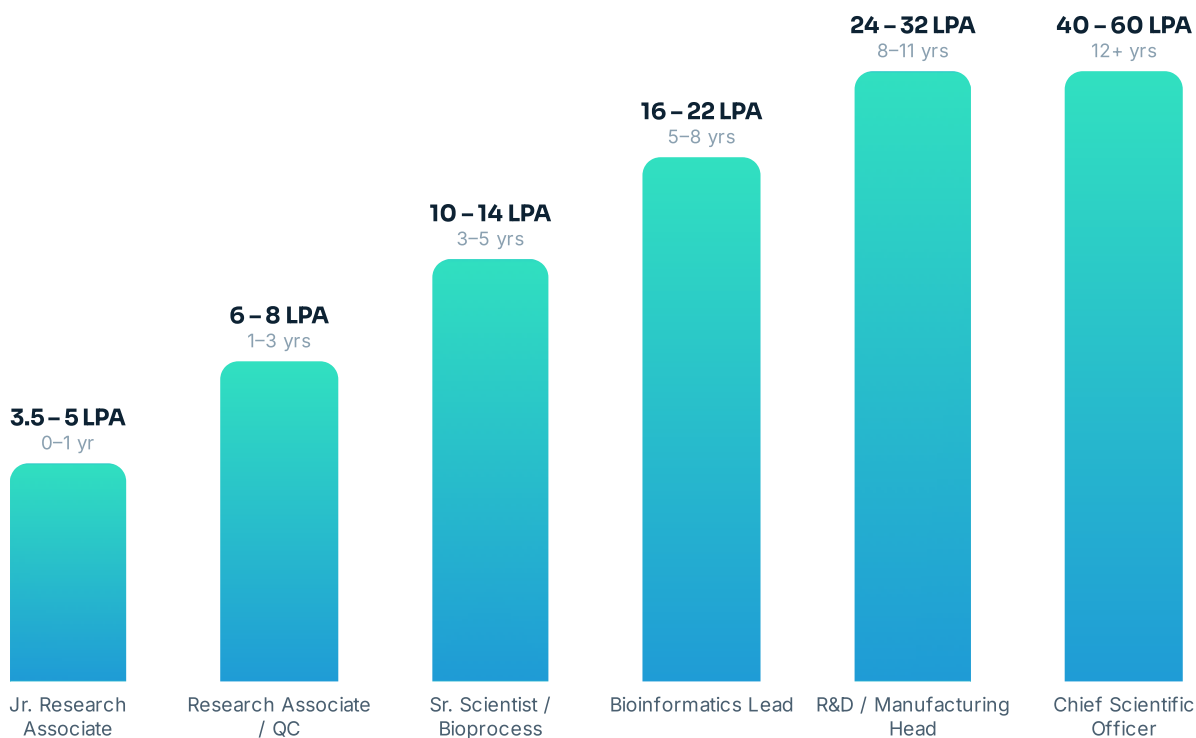
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Electives Per learner Sessions each



GROWTH & COMPENSATION

Your Career Trajectory



Indicative CTC benchmarks for Indian biotech, pharma and CRO organisations, 2026.

SALARY PREMIUM

Add a 25–40% premium with bioinformatics + regulatory dual expertise.

HIRING PARTNERS

Biopharma, CDMOs, agri-biotech, diagnostics and deep-tech startups.

GLOBAL MOBILITY

Skills map directly to EU/US GxP and bioinformatics role requirements.

Roles our alumni move into

Bioprocess Engineer

Computational Biologist

QA/RA Associate

Cell Therapy Specialist

Field Application Scientist

Medical Science Liaison



PRACTICE

Labs, Capstones & Certification

SIGNATURE CAPSTONE PROJECTS

01

Design a CRISPR knockout

Guide design, off-target scoring, validation strategy and a full lab notebook.

02

Scale a fermentation process

From shake flask to 50 L: media, kinetics, yield model and cost sheet.

03

RNA-seq to biomarker

Public dataset to differential expression, pathway enrichment and a report.

04

Build a biosensor circuit

Promoter-reporter design in SBOL with a simulated dose-response curve.

05

De-novo protein binder

Generative design, ProteinMPNN sequences and an MD stability screen.

06

Digital twin of a bioreactor

Hybrid mechanistic-ML model with control simulation and TEA sheet.

ASSESSMENT & CERTIFICATION

- Weekly graded lab reports and data-analysis tasks
- Two proctored theory assessments (mid & final)
- Capstone defence scored by an industry panel
- Verifiable digital certificate with a unique ID

MENTORSHIP

- 1:1 mentor from biopharma or research
- Weekly journal-club and paper reviews
- Live troubleshooting clinics for lab work
- Alumni network and referral pipeline

LAB INFRASTRUCTURE & ACCESS



BSL-1/BSL-2 suite

Laminar flow hoods, CO₂ incubators, autoclave and sterile prep bay.



Molecular bench

Thermocyclers, qPCR, gel docs, nanodrop and microcentrifuges.



Bioprocess floor

2 L–10 L benchtop bioreactors with DO/pH control and TFF skid.



Analytical corner

HPLC, UV-Vis spectrophotometry, ELISA reader and plate handling.



Compute cluster

GPU workstations for AlphaFold, docking and NGS pipelines.



Cell-culture room

Mammalian and microbial culture with cryo-storage and QC logs.

Cohorts start every 6 weeks • Weekday and weekend batches

Hybrid delivery: live online theory + on-campus wet-lab intensives in Bangalore.



Every learner receives industry-recognised, accredited certificates on completion of the internship, industrial training and capstone project modules.

ACCREDITED CERTIFICATES



INTERNSHIP CERTIFICATE

Accredited Certification



INDUSTRIAL TRAINING

Accredited Certification



PROJECT COMPLETION

Accredited Certification

WHAT EACH CREDENTIAL PROVES



Internship

Verified internship completion credential.



Training

Industrial training hours certified.



Project

Capstone project completion proof.

VERIFICATION & ASSESSMENT

Verifiable digital ID

Every certificate carries a unique ID that employers can validate online, along with the graded lab and capstone record behind it.

Assessed, not attended

Credentials are issued only after proctored theory assessments, weekly graded lab reports and a panel-defended capstone.

Three credentials, one programme

Internship, industrial training and capstone certificates are issued together on successful completion.



Learners are prepared for and certified through globally recognised professional certification programs from leading technology companies.

CERTIFICATION TRACKS

**CISCO**

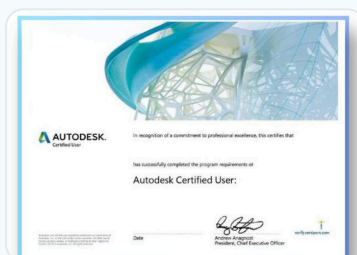
Cybersecurity CCST

**MICROSOFT**

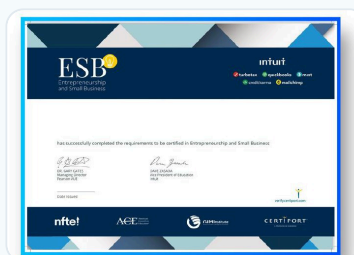
Azure Fundamentals

**IBM**

Internet of Things

**AUTODESK**

Certified User

**ESB**

Entrepreneurship

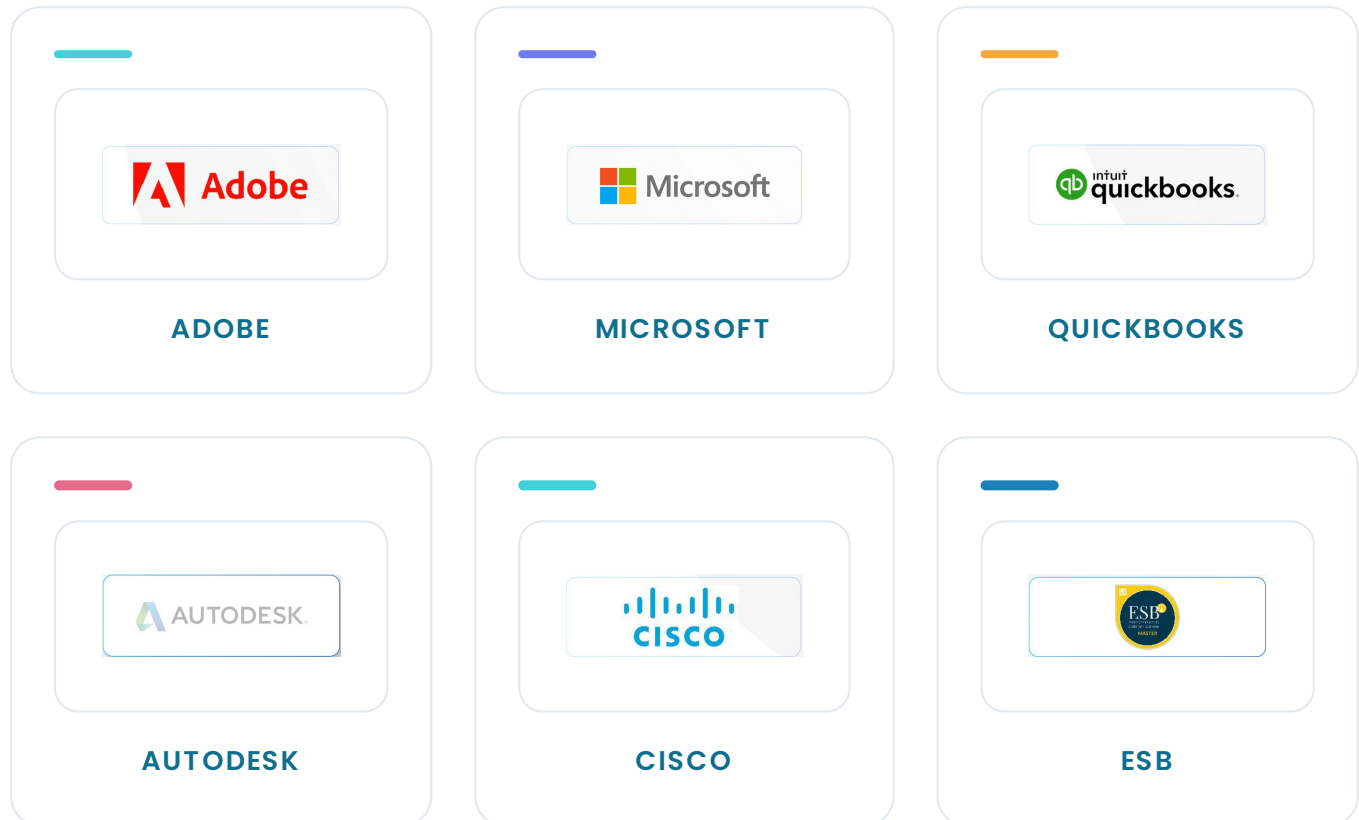
Exam preparation is built into the schedule

Dedicated prep sessions, practice tests and on-campus proctored exams — vouchers guided by your mentor.



Edufyi Tech Solutions is an authorised certification and testing partner for the world's leading technology certification bodies.

PARTNER NETWORK



WHAT PARTNERSHIP MEANS FOR YOU

On-campus exam centre

Sit official vendor exams at our authorised testing centre — no third-party scheduling or travel required.

Original curriculum & tools

Training runs on official vendor courseware and licensed software, delivered by certified instructors.

Authorised Testing & Certification Centre

Certify on campus with globally accepted credentials.



CERTIFY. BUILD. GET HIRED.

CREDENTIALS YOU CAN EARN



Wipro DICE
ID
Accredited



Microsoft
Azure
Fundamentals



Cisco
Cybersecurity
CCST



IBM Internet
of Things



Adobe
Certified
Professional



Autodesk
Certified
User



ESB
Entrepreneurship



QuickBooks
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